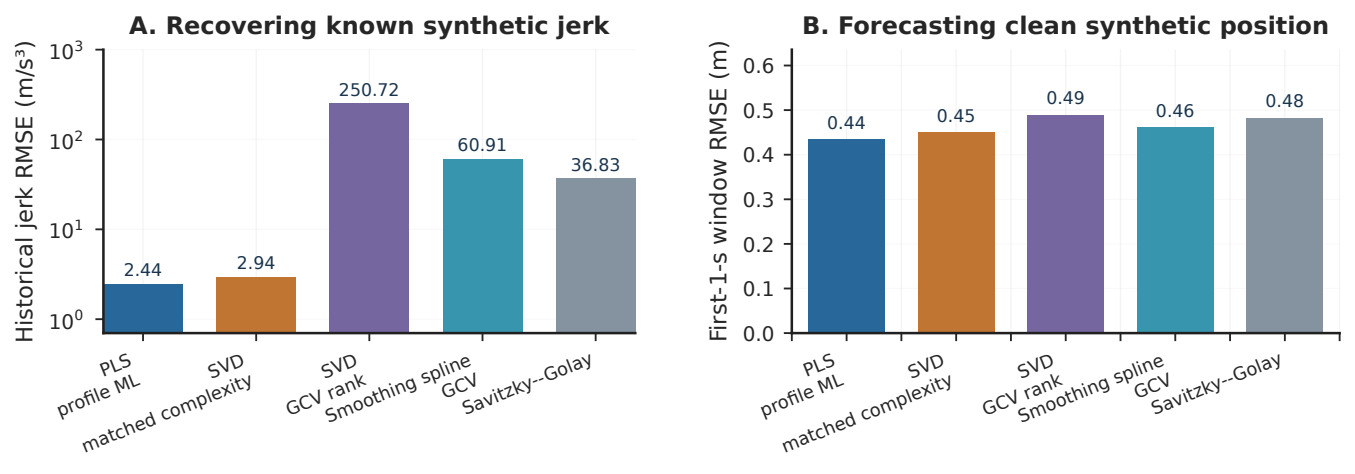
A. Recovering known synthetic jerk
Historical jerk RMSE (m/s³)
10^3
10^2
10^1
10^0
2.44
2.94
250.72
60.91
36.83
PLS profile ML
SVD matched complexity
SVD GCV rank
Smoothing spline GCV
Savitzky--Golay
B. Forecasting clean synthetic position
First-1-s window RMSE (m)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.44
0.45
0.49
0.46
0.48
PLS profile ML
SVD matched complexity
SVD GCV rank
Smoothing spline GCV
Savitzky--Golay